# Curtis McDonald

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## Research Interests

I work on the design and analysis of randomized algorithms for Bayesian computation. For such algorithms I study their computational complexity via mixing time for MCMC and error control for diffusion models, as well as the quality of their output via generalization risk bounds for the estimators they produce. I also study robustness to models and priors in stochastic control.

**Keywords:** MCMC and sampling algorithms; diffusion models; Bayesian computation; PAC-Bayes and generalization; statistical learning theory; stochastic control and filtering.

## Academic Appointments

**Simons Institute, UC Berkeley**, Berkeley, CA — September 2025–
**Machine Learning Pod**
Postdoctoral Researcher
Hosted by Peter Bartlett and Bin Yu

- Deriving an exact score representation theory for reward-tilted densities, enabling a single pretrained diffusion model to sample discrete distributions across a continuum of tilt strengths without retraining, with empirical solution rates matching theory.
- Establishing a split-budget theory for score estimation, distributing a fixed query budget across multiple locations and pooling via ridge regression to recover non-trivial terminal $W_2$ error control even when the base score estimator degrades substantially.
- Contributing statistical and probabilistic analysis to AI safety research, including published work on capability emergence in grokking and ongoing work on diffusion models for adversarial prompting in large language models.

## Education

**Yale University**, New Haven, CT — May 2025
**Ph.D.** in Statistics and Data Science
Advisor: Andrew Barron
Francis J. Anscombe award for outstanding academic performance
Thesis: "Computation and Estimation for Neural Networks via Log-Concave Coupling"

**Queen's University**, Kingston, ON — November 2019
**M.A.Sc.** in Applied Mathematics and Engineering
Advisor: Serdar Yüksel
Thesis: "Filter Stability, Observability and Robustness for Partially Observed Stochastic Dynamical Systems"

**Queen's University**, Kingston, ON — June 2017
**B.A.Sc.** in Applied Mathematics and Engineering
Specialisation in Computing and Communications
Thesis: "Decentralized Learning for Power Grid Control"

## Publications

***** (Selected Publications)

### Pre-Prints

(PP1)***** **McDonald, C.**, Bartlett, P. *"Internal Guidance as a Sampling Method for Tilted Diffusion Models"*. Under Review (2026). [arXiv]

(PP2)***** **McDonald, C.**, Rojas, K., Tao, M., Bartlett, P. *"Improving score-based sampling via affine post-processing"*. Under Review (2026), [arXiv]

(PP3) **McDonald, C.**, "*Technical note on relating scores of tilted distributions*". Preprint (2026), [arXiv]

### Journal Articles

(J1)***** **McDonald, C.**, Barron, A. *"Rapid Bayesian Computation and Estimation for Neural Networks via Log-Concave Coupling"*. Math. Stat. Learn. (2026). published online first. DOI 10.4171/MSL/59 [Link]

(J1a) *Conference version:* **McDonald, C.**, Barron, A. (2024) *"Log-Concave Coupling for Sampling Neural Net Posteriors"*. 2024 IEEE International Symposium on Information Theory (ISIT). pp 2251-2256. [Link]

(J2) **McDonald, C.**, Yüksel, S. *"Stochastic Observability and Filter Stability under Several Criteria"*. IEEE Trans. Automatic Control. vol. 69, no. 5, pp. 2931-2946, May 2024. [Link]

(J2a) *Conference version:* **McDonald, C.**, Yüksel, S. (2018) *"Stability of Non-Linear Filters, Observability and Relative Entropy"*. 2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton). pp 110 - 114. [Link]

(J3) **McDonald, C.**, Yüksel, S. *"Robustness to incorrect priors and controlled filter stability in partially observed stochastic control"*. SIAM Journal on Control and Optimization. vol 60, issue 2 . April 2022. [Link]

(J3a) *Conference version:* **McDonald, C**., Yüksel, S. (2019) *"Observability and Filter Stability for Partially Observed Markov Processes"*. 2019 IEEE 58th Conference on Decision and Control (CDC). pp. 1623-1628. [Link]

(J4) **McDonald, C.**, Yüksel, S. *"Exponential filter stability via Dobrushin's coefficient"*. Electronic Communications in Probability. vol 25, pp. 1-13. August 2020. [Link]

(J5) **McDonald, C**., Alajaji, F., Yüksel, S. *"Two-Way Gaussian Networks with a Jammer and Decentralized Control"*. IEEE Transactions on Control of Network Systems. vol. 7, no. 1, pp. 446-457, March 2020. [Link]

(J5a) *Conference version:* **McDonald, C.**, Alajaji, F., Yüksel, S. (2018) *"Two-Way Gaussian Channels with an Intelligent Jammer"*. 2018 Annual American Control Conference (ACC). pp. 1784-1789. [Link]

### Conference Papers

(C1) Dern, N., Mazioud, S., Heiss, J., Ghosh, A., **McDonald, C.**, Clara, G., Yu, B. (2026) *"Tracking Training Phases in Compositional Learning with Task-Agnostic Measures"*. CompLearn 2026 Workshop at the International Conference on Machine Learning (ICML). [Link]

(C2) **McDonald, C.**, Barron, A. (2022) *"Proposal of a Score Based Approach to Sampling Using Monte Carlo Estimation of Score and Oracle Access to Target Density"*. NeurIPS 2022

Conference Workshop on Score Based Methods. [Link]

## Interdisciplinary Collaborations

(IC1) Sieberer J. M., Park N., Manafzadeh A. R., Desroches S. T., Brennan K., **McDonald C.**, Tommasini S. M., Wiznia D. H., Fulkerson J. P. *"Visualization of Trochlear Dysplasia Using 3-Dimensional Curvature Analysis in Patients With Patellar Instability Facilitates Understanding and Improves the Reliability of the Entry Point to Trochlea Groove Angle"*. Arthroscopy, Sports Medicine, and Rehabilitation. September, 2024. [Link]

(IC2) Sieberer J. M., Park N., Rancu A. L., Desroches S. T., **McDonald C.**, Manafzadeh A. R., Tommasini S. M., Wiznia D. H., Fulkerson J. P. *"Analyzing Alignment Error in Tibial Tuberosity–Trochlear Groove Distance in Clinical Scans Using 2D and 3D Methods"*. The American Journal of Sports Medicine. September, 2024. [Link]

(IC3) Beitler B. G., Sieberer J., Islam W., **McDonald C.**, Yu K., Tommasini S. M., Fulkerson J. P. *"The Morphologic Patella Entry Point Into The Proximal Trochlea Is More Lateral in Recurrent Dislocators Than Controls as Measured by Entry Point-Trochlear Groove Angle"*. Arthroscopy: The Journal of Arthroscopic and Related Surgery. May, 2024. [Link]

(IC4) Kim, J., **McDonald, C.**, Meosky, P., Katsaros, M., Tyler, T. *"Promoting Online Civility Through Platform Architecture"*. Journal of Online Trust and Safety. vol 1, issue 4. September 2022. [Link]

(IC5) Kumar N., Corpus I., Hans M., Harle N., Yang N., **McDonald C.**, Nakamura Sakai S., Janmohamed K., Chen K., Altice F. L., Tang W., Schwartz J. L., Jones-Jang S. M., Saha K., Memon S. A., Bauch C. T., Choudhury M. D., Papakyriakopoulos O., Tucker J. D., Goyal A., Tyagi A., Khoshnood K., Omer S. *"COVID-19 vaccine perceptions in the initial phases of US vaccine roll-out: an observational study on reddit"*. BMC Public Health. vol 22, issue 1, March 2022. [Link]

## Dissertations

- McDonald, Curtis James. *"Computation and Estimation for Neural Networks via Log-Concave Coupling"*. Diss. Yale University (USA), 2025.[Link]
- Mcdonald, Curtis James. *"Filter Stability, Observability and Robustness for Partially Observed Stochastic Dynamical Systems"*. Diss. Queen's University (Canada), 2019. [Link]

## Presentations

### Conference Talks

**ISIT**, Athens, Greece — July 2024
*Log-Concave Coupling for Sampling Neural Net Posteriors*

**CDC**, Nice, France — December 2019
*Observability and Filter Stability for Partially Observed Markov Processes*

**Allerton**, Monticello, IL — October 2018
*Stability of Non-Linear Filters, Observability and Relative Entropy*

**ACC**, Milwaukee, WI — June 2018
*Two-Way Gaussian Channels with an Intelligent Jammer*

### Department Seminars

**Machine Learning Theory Seminar**, UC Berkeley — August 2026
*Tilted Diffusion for Discrete Densities: Exploiting Symmetries of the Hypercube*

**Machine Learning Group Research Talk**, UC Berkeley — November 2025
*SGD and Auxiliary Random Variable MCMC: Similarities and Differences*

**Statistics Student Seminar**, UC Berkeley — September 2025
*Log-Concave Coupling: A Method for Bayesian Estimation and Computation in Single Hidden Layer Neural Networks*

**Meet the Fellows**, Simons Institute, UC Berkeley — September 2025
*Sampling Algorithms for Machine Learning with Auxiliary Random Variables and Diffusion Models*

**Yu Group Meeting**, UC Berkeley — September 2025
*Diffusion Models for Inference*

**Stochastic Control (and Related Fields) Seminar**, Queen's University — September 2024
*Log-Concave Coupling for Rapid Bayesian Neural Network Sampling*

**Theory Students Lunch**, Yale University — February 2024
*Computing Score Based Diffusion Via Log-Concave Sampling*

**Optimization, Sampling, and Games Reading Group**, Yale University — October 2023
*Optimal Transport Gradient Descent Problem*

**Stochastic Control (and Related Fields) Seminar**, Queen's University — March 2023
*Score Based Diffusion as a Density Evolution Method*

**Statistics Student Research Seminar**, Yale University — February 2022
*Markov Chain Monte Carlo and Modern Problems*

**Stochastic Control (and Related Fields) Seminar**, Queen's University — July 2021
*Utilizing the Bakry-Emery Condition in Markov Chain Monte Carlo*

**Statistics Students Research Seminar**, Yale University — July 2020

*A Geometric Approach to Single Layer Neural Nets*

### Posters

**NeurIPS Workshop on Score Based Methods** December 2022
*Proposal of a Score Based Approach to Sampling Using Monte Carlo Estimation of Score and Oracle Access to Target Density*

## Teaching Experience

### University of California, Berkeley

**Lecturer, Department of Statistics** Summer 2026
STAT 20: Introduction to Probability and Statistics — Instructor of Record, 50 students

- Delivered 9 hours per week of in-person content over an 8-week session, covering probability, hypothesis testing, linear regression, and data summary and visualization in R.
- Taught in the department's flipped-classroom format, with live coding demonstrations, group problem solving, and in-class work sessions with active instructor feedback.
- Rewrote the three-lecture series on introduction to probability theory, restructuring axioms and set theory, uniform densities and combinatorics, and pmf's and density functions into a single coherent arc.
- Wrote all in-class examinations; managed the course website, assignment schedule, and grade administration; supervised and coordinated a staff of 6 tutors and graders.
- Taught a cohort spanning first-year, transfer, visiting international, and concurrently enrolled high school students, supporting widely varying quantitative preparation through office hours and individual follow-up with struggling students.

### Yale University

**Teaching Fellow, Department of Statistics and Data Science** 2020–2025

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| S&DS 364 | Information Theory | Spring 2025 |
| S&DS 238 | Probability and Bayesian Statistics | Fall 2024 |
| S&DS 312 | Linear Models | Fall 2021, Fall 2022 |
| S&DS 351 | Stochastic Processes | Spring 2022 |
| S&DS 365 | Applied Data Mining and Machine Learning | Spring 2021 |
| S&DS 242 | Theory of Statistics | Spring 2020 |

- Led weekly discussion sections and office hours across 7 course offerings spanning introductory theory, applied machine learning, and graduate probability.
- Graded problem sets and contributed to problem set design and course materials.
- Supported the department's transition to online and hybrid instruction (2020–2021).

### Queen's University

**Teaching Assistant, Faculty of Engineering and Applied Science** 2015–2019

- Led tutorials and graded for undergraduate courses in calculus, linear algebra, functional analysis, and complex analysis.
- Delivered all lectures and coordinated instruction for a 300-student calculus course during a two-week instructor leave of absence.

## Fellowships and Awards

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| Francis J. Anscombe Award | Yale University | Academic Award | 2025 |
| NSERC Postgraduate Scholarship – Doctoral | Yale University | $63,000 | 2019–2021 |
| Nominee, Teaching Assistant Excellence Award | Queen's University | Teaching Award | 2018 |
| Alexander G. Bell Canada Graduate Scholarship | Queen's University | $17,500 | 2018 |
| Queen Elizabeth II Scholarship in Science and Technology | Queen's University | $15,000 | 2017 |
| NSERC Undergraduate Summer Research Award | Queen's University | $4,500 | 2016–2017 |

## Professional Service

### Review Duties

#### Grant Review

German Research Foundation (DFG) 2026

#### Journal Refereeing

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|---|---|
| IEEE Transactions on Automatic Control | 2024-2026 |
| IEEE Transactions on Control of Network Systems | 2023, 2024 |
| Mathematics of Operations Research | 2024, 2026 |
| Foundations of Data Science | 2021 |

#### Conference Refereeing

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|---|---|
| IEEE Conference on Decision and Control (CDC) | 2025, 2026 |
| American Control Conference (ACC) | 2024, 2026 |
| IEEE International Symposium on Information Theory (ISIT) | 2024, 2025 |

### Service

**Founder and Facilitator**, Mathematical Foundations of Diffusion Models Reading Group Fall 2026
*Simons Institute, UC Berkeley*

**Organizer**, Machine Learning Theory Reading Group — Spring, Fall 2026
*Simons Institute, UC Berkeley*

**Organizing Committee**, Workshop Honoring Andrew Barron: Forty Years at the Interplay of Information Theory, Probability and Statistical Learning — April 2024
*Yale Institute for Foundations of Data Science*

**Poster Selection Committee**, Recent Advances and Future Directions for Sampling — October 2024
*Yale Institute for Foundations of Data Science*

**Graduate Recruitment Host**, Admitted Students Visit Days — 2021–2024
*Department of Statistics and Data Science, Yale University*

**Student Social Organizer** — 2020–2025
*Department of Statistics and Data Science, Yale University*